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Quaternion Windowed Linear Canonical Transform of Two-Dimensional Signals

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Abstract. We investigate the 2D quaternion windowed linear canonical transform (QWLCT) in this paper. Firstly, we propose the new definition of the QWLCT, and then several important properties of newly defined QWLCT, such as bounded, shift, modulation, orthogonality relation, are derived based on the spectral representation of the quaternionic linear canonical transform (QLCT). Secondly, by the Heisenberg uncertainty principle for the QLCT and the orthogonality relation property for the QWLCT, the Heisenberg uncertainty principle for the QWLCT is es- tablished. Finally, we give an example of the QWLCT.
Keywords. Quaternion linear canonical transform, Quaternion windowed linear canonical transform, Uncertainty principle.

1. Introduction

It is well-known that the linear canonical transform (LCT) has found broad applications in many fields of applied mathematics, signal processing, radar system analysis, filter design, phase retrieval, and pattern recognition and optics [1,6,7,11]. All of these results show that the LCT plays an important role for chirp signal analysis in parameter estimation, sampling and filtering [6–8,11,13,16,21,22,24,26,30]. However, the LCT can't show the local LCT-frequency contents as a result of its global kernel. In Refs. [16,17] the LCT has been successfully applied to research the generalized windowed function. The windowed linear canonical transform (WLCT) is a method devised to study signals whose spectral content changes in time. As shown in Refs. [4,16] some important properties of the WLCT are discussed. Those include the analogue of the Poisson summation formula, sampling formulas, Paley–Wiener theorem, and uncertainly relations. It presents the time and LCT-frequency information, and is originally a local LCT distribution.

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On the other hand, some authors have generalized the linear canonical transform to quaternion-valued signals, known as the quaternionic linear canonical transform (QLCT). The QLCT was firstly studied in Ref. [23] including prolate spheroidal wave signals and uncertainty principles [27]. Some useful properties of the QLCT such as linearity, reconstruction formula, continuity, boundedness, positivity inversion formula and the uncertainty principle were established in Refs. [12,15,18,19,23,33]. An application of the QLCT to study of generalized swept-frequency filters was introduced in Ref. [2]. Because of the non-commutative property of multiplication of quaternions, there are mainly three various types of 2D quaternion linear canonical transform (QLCTs): two-sided QLCTs, left-sided QLCTs and right-sided QLCTs (refer to [14]). Based on the (two-sided) QLCT, one may extend the WLCT to quaternion algebra while possessing similar properties as in the classical case.

The WLCT has been widely used, but the quaternion windowed linear canonical transform (QWLCT) hasn't been mentioned yet. The main goals of the present paper are to study the properties of the QLCT of 2D quaternionic signals and to derive the novel concept of the QWLCT, and research important properties of the QWLCT such as boundedness, shift, modulation, orthogonality relation. Using the Heisenberg uncertainty principle for the QLCT and the orthogonality relation property for the QWLCT, we establish a generalized QWLCT uncertainty principle. This also lays a good foundation for the practical application in the future.

The paper is organized as follows: Sect. 2 gives a brief introduction to some general definitions and basic properties of quaternion algebra, QLCTs of 2D Quaternion-valued signals. We give the definition and study the properties of the QWLCT in Sect. 3. In Sect. 4, provides the uncertainty principle associated with the QWLCT. We give an example of the QWLCT in Sect. 5. In Sect. 6, some conclusions are drawn.

2. Preliminary

In this section, we mainly review some basic facts on the quaternion algebra and the QLCT, which will be needed throughout the paper.

2.1. Quaternion Algebra

The quaternion algebra is an extension of the complex number to 4D algebra. It was first invented by W. R. Hamilton in 1843 and classically denoted by $\mathbb{H}$ in his honor. Every element of $\mathbb{H}$ has a Cartesian form given by

$$\mathbb{H} = \{q | q := [q]_0 + \mathbf{i}[q]_1 + \mathbf{j}[q]_2 + \mathbf{k}[q]_3, [q]_i \in \mathbb{R}, \quad i = 0, 1, 2, 3\}, \tag{2.1}$$

where $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are imaginary units obeying Hamilton's multiplication rules (see [14])

$$\begin{aligned} &\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1, \\ &\mathbf{ij} = -\mathbf{ji} = \mathbf{k}, \ \mathbf{jk} = -\mathbf{kj} = \mathbf{i}, \ \mathbf{ki} = -\mathbf{ik} = \mathbf{j}. \end{aligned} \tag{2.2}$$

Let $[q]_0$ and $q = \mathbf{i}[q]_1 + \mathbf{j}[q]_2 + \mathbf{k}[q]_3$ denote the real scalar part and the vector part of quaternion number $q = [q]_0 + \mathbf{i}[q]_1 + \mathbf{j}[q]_2 + \mathbf{k}[q]_3$, respectively. Then, from [9],the real scalar part has a cyclic multiplication symmetry

$$[pql]_0 = [qlp]_0 = [lpq]_0, \quad \forall q, p, l \in \mathbb{H}, \tag{2.3}$$

the conjugate of a quaternion q is defined by $\overline{q} = [q]_0 - \mathbf{i}[q]_1 - \mathbf{j}[q]_2 - \mathbf{k}[q]_3$, and the norm of $q \in \mathbb{H}$ defined as

$$|q| = \sqrt{q\overline{q}} = \sqrt{[q]_0^2 + [q]_1^2 + [q]_2^2 + [q]_3^2}. \tag{2.4}$$

It is easy to verify that

$$\overline{pq} = \overline{q}\ \overline{p}, |qp| = |q||p|, \quad \forall q, p \in \mathbb{H}. \tag{2.5}$$

The quaternion modules $L^2(\mathbb{R}^2, \mathbb{H})$ are defined as

$$L^2(\mathbb{R}^2, \mathbb{H}) := \{f | f : \mathbb{R}^2 \to \mathbb{H}, \|f\|_{L^2(\mathbb{R}^2,\mathbb{H})} = \int_{\mathbb{R}^2} |f(x_1, x_2)|^2 \mathrm{d}x_1 \mathrm{d}x_2 < \infty\}. \tag{2.6}$$

Now we introduce an inner product of quaternion functions f, g defined on $L^2(\mathbb{R}^2, \mathbb{H})$ given by

$$(f, g)_{L^2(\mathbb{R}^2,\mathbb{H})} = \int_{\mathbb{R}^2} f(\mathbf{x})\overline{g(\mathbf{x})}\mathrm{d}\mathbf{x}, \tag{2.7}$$

with symmetric real scalar part

$$\langle f, g\rangle = \frac{1}{2}\{(f, g) + (g, f)\} = \int_{\mathbb{R}^2} [f(\mathbf{x})\overline{g(\mathbf{x})}]_0 \mathrm{d}\mathbf{x}, \tag{2.8}$$

where $\mathbf{x} = (x_1, x_2) \in \mathbb{R}^2$ and $\mathrm{d}\mathbf{x} = \mathrm{d}x_1\mathrm{d}x_2$.

The associated scalar norm of $f(\mathbf{x}) \in L^2(\mathbb{R}^2, \mathbb{H})$ is defined by both (2.7) and (2.8):

$$\|f\|^2_{L^2(\mathbb{R}^2,\mathbb{H})} = \langle f, f\rangle_{L^2(\mathbb{R}^2,\mathbb{H})} = \int_{\mathbb{R}^2} |f(\mathbf{x})|^2 \mathrm{d}\mathbf{x} < \infty. \tag{2.9}$$

Lemma 2.1. *If $f, g \in L^2(\mathbb{R}^2, \mathbb{H})$, then the Cauchy–Schwarz inequality holds[9]*

$$\left|\langle f, g\rangle_{L^2(\mathbb{R}^2,\mathbb{H})}\right|^2 \leq \|f\|^2_{L^2(\mathbb{R}^2,\mathbb{H})} \|g\|^2_{L^2(\mathbb{R}^2,\mathbb{H})}. \tag{2.10}$$

If and only if $f = \lambda g$ for some quaternionic parameter $\lambda \in \mathbb{H}$, the equality holds.

2.2. The Quaternion Linear Canonical Transform

The QLCT is firstly defined by Kou et al., which is a generalization of the LCT in the frame of quaternion algebra [14]. Due to the non-commutativity of quaternion multiplication, there are three different types of the QLCT: the left-sided QLCT, the right-sided QLCT, and the two-sided QLCT. In this paper, we mainly focus on the two-sided QLCT.

Definition 2.2. Let $A_i = \begin{bmatrix} a_i & b_i \\ c_i & d_i \end{bmatrix} \in \mathbb{R}^{2\times 2}$ be a matrix parameter satisfying $\det(A_i) = 1$, for $i = 1, 2$. The two-sided QLCT of signal $f \in L^2\left(\mathbb{R}^2, \mathbb{H}\right)$ is defined by

$$\mathcal{L}^{\mathbb{H}}_{A_1,A_2}\{f\}(\mathbf{w}) = \int_{\mathbb{R}^2} K^{\mathbf{i}}_{A_1}(x_1,\omega_1) f(\mathbf{x}) K^{\mathbf{j}}_{A_2}(x_2,\omega_2) \mathrm{d}\mathbf{x}, \tag{2.11}$$

where $\mathbf{w} = (\omega_1, \omega_2) \in \mathbb{R}^2$ is regarded as the QLCT domain, and the kernel signals $K^{\mathbf{i}}_{A_1}(x_1,\omega_1)$, $K^{\mathbf{j}}_{A_2}(x_2,\omega_2)$ are respectively given by

$$K^{\mathbf{i}}_{A_1}(x_1,\omega_1) := \begin{cases} \frac{1}{\sqrt{2\pi b_1}} e^{\mathbf{i}\left(\frac{a_1}{2b_1}x_1^2 - \frac{x_1\omega_1}{b_1} + \frac{d_1}{2b_1}\omega_1^2 - \frac{\pi}{4}\right)}, & b_1 \neq 0 \\ \sqrt{d_1} e^{\mathbf{i}\frac{c_1 d_1}{2}\omega_1^2} \delta(x_1 - d_1 w_1), & b_1 = 0 \end{cases} \tag{2.12}$$

and

$$K^{\mathbf{j}}_{A_2}(x_2,\omega_2) := \begin{cases} \frac{1}{\sqrt{2\pi b_2}} e^{\mathbf{j}\left(\frac{a_2}{2b_2}x_2^2 - \frac{x_2\omega_2}{b_2} + \frac{d_2}{2b_2}\omega_2^2 - \frac{\pi}{4}\right)}, & b_2 \neq 0 \\ \sqrt{d_2} e^{\mathbf{j}\frac{c_2 d_2}{2}\omega_2^2} \delta(x_2 - d_2 w_2), & b_2 = 0, \end{cases} \tag{2.13}$$

where $\delta(x)$ representing the Dirac function.

From the above definition, it is noted that for $b_i = 0, i = 1, 2$ the QLCT of a signal is essentially a chirp multiplication and it is of no particular interest for our objective in this work. Hence, without loss of generality, we set $b_i \neq 0$ in the following section unless stated otherwise. Under some suitable conditions, the QLCT above is invertible and the inversion is given in the following section.

Definition 2.3. Suppose $f \in L^2\left(\mathbb{R}^2, \mathbb{H}\right)$, then the inversion of the QLCT of f is given by

$$\begin{aligned} f(\mathbf{x}) &= \mathcal{L}^{-1}_{A_1,A_2}[\mathcal{L}^{\mathbb{H}}_{A_1,A_2}\{f\}](\mathbf{x}) \\ &= \int_{\mathbb{R}^2} K^{-\mathbf{i}}_{A_1}(x_1,\omega_1) \mathcal{L}^{\mathbb{H}}_{A_1,A_2}\{f\}(\mathbf{w}) K^{-\mathbf{j}}_{A_2}(x_2,\omega_2) \mathrm{d}\mathbf{w}. \end{aligned} \tag{2.14}$$

The Parseval formula of QLCT is very available as follows:

Lemma 2.4. (QLCT Parseval) *[23] Two quaternion functions $f, g \in L^2(\mathbb{R}^2, \mathbb{H})$ are related to their QLCT via the Parseval formula, given as*

$$\langle f, g\rangle_{L^2(\mathbb{R}^2,\mathbb{H})} = \langle \mathcal{L}^{\mathbb{H}}_{A_1,A_2}\{f\}, \mathcal{L}^{\mathbb{H}}_{A_1,A_2}\{g\}\rangle_{L^2(\mathbb{R}^2,\mathbb{H})}. \tag{2.15}$$

For $f = g$, one have

$$\|f\|^2_{L^2(\mathbb{R}^2,\mathbb{H})} = \|\mathcal{L}^{\mathbb{H}}_{A_1,A_2}\{f\}\|^2_{L^2(\mathbb{R}^2,\mathbb{H})} \tag{2.16}$$

3. Quaternionic Windowed Linear Canonical Transform (QWLCT)

In this section, the generalization of window function associated with the QLCT will be discussed, which is denoted by QWLCT. Moreover, several basic properties of them are investigated.

3.1. The Definition of 2D QWLCT

This section leads to 2D quaternion window function associated with the QLCT. Due to the noncommutative property of multiplication of quaternions, there are three different types of QWLCTs: two-sided QWLCT, left-sided QWLCT and right-sided QWLCT. Alternatively, we use the (Two-sided) QWLCT to define the QWLCT.

Definition 3.1. (QWLCT). Let $A_i = \begin{bmatrix} a_i & b_i \\ c_i & d_i \end{bmatrix} \in \mathbb{R}^{2\times 2}$ be a matrix parameter satisfying $\det(A_i) = 1$, for $i = 1, 2$. Let $\phi \in L^2(\mathbb{R}^2, \mathbb{H})\backslash\{0\}$ be a quaternion window function. The two-sided QWLCT of a signal $f \in L^2\left(\mathbb{R}^2, \mathbb{H}\right)$ with respect to ϕ is defined by

$$G_\phi^{A_1,A_2} f(\mathbf{w}, \mathbf{u}) = \int_{\mathbb{R}^2} K_{A_1}^{\mathbf{i}}(x_1, \omega_1) f(\mathbf{x}) \overline{\phi(\mathbf{x}-\mathbf{u})} K_{A_2}^{\mathbf{j}}(x_2, \omega_2) \mathrm{d}\mathbf{x}, \tag{3.1}$$

where $\mathbf{u} = (u_1, u_2) \in \mathbb{R}^2$, $K_{A_1}^{\mathbf{i}}(x_1, \omega_1)$ and $K_{A_2}^{\mathbf{j}}(x_2, \omega_2)$ are given by (2.12) and (2.13), respectively.

For a fixed $\mathbf{u}$, we have

$$G_\phi^{A_1,A_2} f(\mathbf{w}, \mathbf{u}) = \mathcal{L}_{A_1,A_2}^{\mathbb{H}}\{f(\mathbf{x}) \overline{\phi(\mathbf{x}-\mathbf{u})}\}(\mathbf{w}). \tag{3.2}$$

Applying the inverse QLCT to (3.2), we have

$$\begin{aligned} f(\mathbf{x}) \overline{\phi(\mathbf{x}-\mathbf{u})} &= \mathcal{L}_{A_1,A_2}^{-1}\{G_\phi^{A_1,A_2} f(\mathbf{w}, \mathbf{u})\} \\ &= \int_{\mathbb{R}^2} K_{A_1}^{-\mathbf{i}(x_1,\omega_1)} G_\phi^{A_1,A_2}\{f\}(\mathbf{w}, \mathbf{u}) K_{A_2}^{-\mathbf{j}(x_2,\omega_2)} \mathrm{d}\mathbf{w}. \end{aligned} \tag{3.3}$$

3.2. Some Properties of QWLCT

In this subsection, we discuss several basic properties of the QWLCT. These properties play important roles in signal representation.

Theorem 3.2. (Boundedness) *Let $\phi \in L^2(\mathbb{R}^2, \mathbb{H})\backslash\{0\}$ be a window function and $f \in L^2\left(\mathbb{R}^2, \mathbb{H}\right)$, then*

$$|G_\phi^{A_1,A_2} f(\mathbf{w}, \mathbf{u})| \le \frac{1}{2\pi\sqrt{|b_1 b_2|}} \|f\|_{L^2(\mathbb{R}^2)} \|\phi\|_{L^2(\mathbb{R}^2)}. \tag{3.4}$$

Proof. By using the quaternion Cauchy-Schwarz inequality (2.10)

$$\begin{aligned} |G_\phi^{A_1,A_2} f(\mathbf{w}, \mathbf{u})|^2 &= \left| \int_{\mathbb{R}^2} K_{A_1}^{\mathbf{i}}(x_1, \omega_1) f(\mathbf{x}) \overline{\phi(\mathbf{x}-\mathbf{u})} K_{A_2}^{\mathbf{j}}(x_2, \omega_2) \mathrm{d}\mathbf{x} \right|^2 \\ &\le \left(\int_{\mathbb{R}^2} \left| K_{A_1}^{\mathbf{i}}(x_1, \omega_1) f(\mathbf{x}) \overline{\phi(\mathbf{x}-\mathbf{u})} K_{A_2}^{\mathbf{j}}(x_2, \omega_2) \right| \mathrm{d}\mathbf{x} \right)^2 \\ &= \left(\frac{1}{\sqrt{4\pi^2 |b_1 b_2|}} \int_{\mathbb{R}^2} \left| f(\mathbf{x}) \overline{\phi(\mathbf{x}-\mathbf{u})} \right| \mathrm{d}\mathbf{x} \right)^2 \\ &\le \frac{1}{4\pi^2 |b_1 b_2|} \left(\int_{\mathbb{R}^2} |f(\mathbf{x})|^2 \, \mathrm{d}\mathbf{x} \right) \left(\int_{\mathbb{R}^2} \left| \overline{\phi(\mathbf{x}-\mathbf{u})} \right|^2 \mathrm{d}\mathbf{x} \right) \\ &= \frac{1}{4\pi^2 |b_1 b_2|} \|f\|_{L^2(\mathbb{R}^2)}^2 \|\phi\|_{L^2(\mathbb{R}^2)}^2, \end{aligned} \tag{3.5}$$

where applying the change of variables $\mathbf{x}-\mathbf{u}=\mathbf{t}$ in the last step. Then we have

$$|G_{\phi}^{A_1,A_2}f(\mathbf{w},\mathbf{u})| \leq \frac{1}{2\pi\sqrt{|b_1b_2|}}\|f\|_{L^2(\mathbb{R}^2)}\|\phi\|_{L^2(\mathbb{R}^2)}, \tag{3.6}$$

which completes the proof. □

Theorem 3.3. (Linearity) *Let $\phi \in L^2(\mathbb{R}^2,\mathbb{H})\backslash\{0\}$ be a window function and $f,g \in L^2\left(\mathbb{R}^2,\mathbb{H}\right)$. The QWLCT is a linear operator, namely,*

$$[G_{\phi}^{A_1,A_2}(\lambda f+\mu g)](\mathbf{w},\mathbf{u}) = \lambda G_{\phi}^{A_1,A_2}f(\mathbf{w},\mathbf{u}) + \mu G_{\phi}^{A_1,A_2}g(\mathbf{w},\mathbf{u}) \tag{3.7}$$

for arbitrary real constants λ and μ .

Proof. This follows directly from the linearity of the product and the integration involved in Definition 3.1. □

Theorem 3.4. (Parity) *Let $\phi \in L^2(\mathbb{R}^2,\mathbb{H})\backslash\{0\}$ be a window function and $f \in L^2\left(\mathbb{R}^2,\mathbb{H}\right)$. Then we have*

$$G_{P\phi}^{A_1,A_2}\{Pf\}(\mathbf{w},\mathbf{u}) = G_{\phi}^{A_1,A_2}f(-\mathbf{w},-\mathbf{u}), \tag{3.8}$$

where $P\phi(\mathbf{x}) = \phi(-\mathbf{x})$ for every window function $\phi \in L^2(\mathbb{R}^2)$.

Proof. A direct calculation gives, for every $f \in L^2\left(\mathbb{R}^2,\mathbb{H}\right)$,

$$\begin{aligned} G_{P\phi}^{A_1,A_2}\{Pf\}(\mathbf{w},\mathbf{u}) &= \int_{\mathbb{R}^2} K_{A_1}^{\mathbf{i}}(x_1,\omega_1)f(-\mathbf{x})\overline{\phi(-(\mathbf{x}-\mathbf{u}))}K_{A_2}^{\mathbf{j}}(x_2,\omega_2)\mathrm{d}\mathbf{x} \\ &= \int_{\mathbb{R}^2} \frac{1}{\sqrt{2\pi b_1}} e^{\mathbf{i}\left(\frac{a_1}{2b_1}(-x_1)^2-\frac{(-x_1)(-\omega_1)}{b_1}+\frac{d_1}{2b_1}(-\omega_1)^2-\frac{\pi}{4}\right)} \\ &\quad \times f(-\mathbf{x})\overline{\phi(-(\mathbf{x}-\mathbf{u}))} \\ &\quad \times \frac{1}{\sqrt{2\pi b_2}} e^{\mathbf{j}\left(\frac{a_2}{2b_2}(-x_2)^2-\frac{(-x_2)(-\omega_2)}{b_2}+\frac{d_2}{2b_2}(-\omega_2)^2-\frac{\pi}{4}\right)}\mathrm{d}\mathbf{x}, \end{aligned} \tag{3.9}$$

which proves the theorem according to Definition 3.1. □

Theorem 3.5. (Shift) *Let $\phi \in L^2(\mathbb{R}^2,\mathbb{H})\backslash\{0\}$ be a window function and $f \in L^2\left(\mathbb{R}^2,\mathbb{H}\right)$. Then we have*

$$G_{\phi}^{A_1,A_2}\{T_{\mathbf{r}}f\}(\mathbf{w},\mathbf{u}) = e^{\mathbf{i}r_1\omega_1c_1}e^{-\mathbf{i}\frac{a_1r_1^2}{2}c_1}G_{\phi}^{A_1,A_2}\{f\}(\mathbf{m},\mathbf{n})e^{\mathbf{j}r_2\omega_2c_2}e^{-\mathbf{j}\frac{a_2r_2^2}{2}c_2}, \tag{3.10}$$

where $T_{\mathbf{r}}f(\mathbf{x}) = f(\mathbf{x}-\mathbf{r}), \mathbf{r} = (r_1,r_2)$, $\mathbf{m} = (m_1,m_2)$, $\mathbf{n} = (n_1,n_2) \in \mathbb{R}^2$, $m_i = w_i - a_ir_i$, $n_i = u_i - r_i$, $i = 1,2$.

Proof. By Definition 3.1, we have

$$G_{\phi}^{A_1,A_2}\{T_{\mathbf{r}}f\}(\mathbf{w},\mathbf{u}) = \int_{\mathbb{R}^2} K_{A_1}^{\mathbf{i}}(x_1,\omega_1)f(\mathbf{x}-\mathbf{r})\overline{\phi(\mathbf{x}-\mathbf{u})}K_{A_2}^{\mathbf{j}}(x_2,\omega_2)\mathrm{d}\mathbf{x}. \tag{3.11}$$

By making the change of variable $\mathbf{t} = \mathbf{x} - \mathbf{r}$ in the above expression, we obtain

$$\begin{aligned}
&G_\phi^{A_1,A_2}\{T_\mathbf{r} f\}(\mathbf{w},\mathbf{u})\\
&= \int_{\mathbb{R}^2} \frac{1}{\sqrt{2\pi b_1}} e^{\mathbf{i}\left(\frac{a_1}{2b_1}(r_1+t_1)^2 - \frac{(r_1+t_1)\omega_1}{b_1} + \frac{d_1}{2b_1}\omega_1^2 - \frac{\pi}{4}\right)} f(\mathbf{t})\overline{\phi(\mathbf{t}-(\mathbf{u}-\mathbf{r}))}\\
&\quad\times \frac{1}{\sqrt{2\pi b_2}} e^{\mathbf{j}\left(\frac{a_2}{2b_2}(r_2+t_2)^2 - \frac{(r_2+t_2)\omega_2}{b_2} + \frac{d_2}{2b_2}\omega_2^2 - \frac{\pi}{4}\right)} \mathrm{d}\mathbf{t}\\
&= \frac{1}{\sqrt{2\pi b_1}} e^{\mathbf{i}\frac{a_1}{2b_1}r_1^2 - \mathbf{i}\frac{r_1\omega_1}{b_1}} \int_{\mathbb{R}^2} e^{\mathbf{i}\left(\frac{a_1}{2b_1}t_1^2 - \frac{(\omega_1 - r_1a_1)t_1}{b_1} + \frac{d_1}{2b_1}(\omega_1 - r_1a_1 + r_1a_1)^2 - \frac{\pi}{4}\right)}\\
&\quad\times f(\mathbf{t})\overline{\phi(\mathbf{t}-(\mathbf{u}-\mathbf{r}))}\frac{1}{\sqrt{2\pi b_2}} e^{\mathbf{j}\frac{a_2}{2b_2}r_2^2 - \mathbf{j}\frac{r_2\omega_2}{b_2}} \mathrm{d}\mathbf{t}\\
&\quad\times e^{\mathbf{j}\left(\frac{a_2}{2b_2}t_2^2 - \frac{(\omega_2 - r_2a_2)t_2}{b_2} + \frac{d_2}{2b_2}(\omega_2 - r_2a_2 + r_2a_2)^2 - \frac{\pi}{4}\right)}\\
&= e^{\mathbf{i}\frac{d_1}{2b_1}(2r_1a_1(\omega_1 - r_1a_1) + (r_1a_1)^2)} e^{\mathbf{i}\frac{a_1}{2b_1}r_1^2} e^{-\mathbf{i}\frac{r_1\omega_1}{b_1}}\\
&\quad\times \int_{\mathbb{R}^2} \frac{1}{\sqrt{2\pi b_1}} e^{\mathbf{i}\left(\frac{a_1}{2b_1}t_1^2 - \frac{t_1(\omega_1 - r_1a_1)}{b_1} + \frac{d_1}{2b_1}(\omega_1 - r_1a_1)^2 - \frac{\pi}{4}\right)} f(\mathbf{t})\overline{\phi(\mathbf{t}-(\mathbf{u}-\mathbf{r}))}\\
&\quad\times \frac{1}{\sqrt{2\pi b_2}} e^{\mathbf{j}\left(\frac{a_2}{2b_2}t_2^2 - \frac{t_2(\omega_2 - r_2a_2)}{b_2} + \frac{d_2}{2b_2}(\omega_2 - r_2a_2)^2 - \frac{\pi}{4}\right)} \mathrm{d}\mathbf{t}\\
&\quad\times e^{\mathbf{j}\frac{d_2}{2b_2}(2r_2a_2(\omega_2 - r_2a_2) + (r_2a_2)^2)} e^{\mathbf{j}\frac{a_2}{2b_2}r_2^2} e^{-\mathbf{j}\frac{r_2\omega_2}{b_2}}.
\end{aligned}\tag{3.12}$$

Applying the definition of the QWLCT, the above expression can be rewritten in the form

$$\begin{aligned}
G_\phi^{A_1,A_2}\{T_\mathbf{r} f\}(\mathbf{w},\mathbf{u}) &= e^{\mathbf{i}\frac{d_1}{2b_1}(2r_1a_1(\omega_1 - r_1a_1) + (r_1a_1)^2)} e^{\mathbf{i}\frac{a_1}{2b_1}r_1^2} e^{-\mathbf{i}\frac{r_1\omega_1}{b_1}}\\
&\quad\times G_\phi^{A_1,A_2}\{f\}(\mathbf{m},\mathbf{n})\\
&\quad\times e^{\mathbf{j}\frac{d_2}{2b_2}(2r_2a_2(\omega_2 - r_2a_2) + (r_2a_2)^2)} e^{\mathbf{j}\frac{a_2}{2b_2}r_2^2} e^{-\mathbf{j}\frac{r_2\omega_2}{b_2}}.
\end{aligned}\tag{3.13}$$

Because $a_id_i - b_ic_i = 1$, for $i = 1, 2$. We get

$$e^{\mathbf{i}\frac{d_1}{2b_1}(2r_1a_1(\omega_1 - r_1a_1) + (r_1a_1)^2)} e^{\mathbf{i}\frac{a_1}{2b_1}r_1^2} e^{-\mathbf{i}\frac{r_1\omega_1}{b_1}} = e^{\mathbf{i}r_1\omega_1c_1} e^{-\mathbf{i}\frac{a_1r_1^2}{2}c_1}, \tag{3.14}$$

$$e^{\mathbf{j}\frac{d_2}{2b_2}(2r_2a_2(\omega_2 - r_2a_2) + (r_2a_2)^2)} e^{\mathbf{j}\frac{a_2}{2b_2}r_2^2} e^{-\mathbf{j}\frac{r_2\omega_2}{b_2}} = e^{\mathbf{j}r_2\omega_2c_2} e^{-\mathbf{j}\frac{a_2r_2^2}{2}c_2}. \tag{3.15}$$

We finally arrive at

$$\begin{aligned}
G_\phi^{A_1,A_2}\{T_\mathbf{r} f\}(\mathbf{w},\mathbf{u}) &= e^{\mathbf{i}r_1\omega_1c_1} e^{-\mathbf{i}\frac{a_1r_1^2}{2}c_1} G_\phi^{A_1,A_2}\{f\}(\mathbf{m},\mathbf{n})\\
&\quad\times e^{\mathbf{j}r_2\omega_2c_2} e^{-\mathbf{j}\frac{a_2r_2^2}{2}c_2},
\end{aligned}\tag{3.16}$$

which completes the proof. □

Theorem 3.6. (Modulation) *Let* $\phi \in L^2(\mathbb{R}^2, \mathbb{H})\backslash\{0\}$ *be a window function and* $f \in L^2(\mathbb{R}^2, \mathbb{H})$. $\mathbb{M}_s f$ *be modulation operator defined by* $\mathbb{M}_s f(\mathbf{x}) =$

$e^{\mathbf{i}x_1 s_1} f(\mathbf{x}) e^{\mathbf{j}x_2 s_2}$ *with* $\mathbf{s}=(s_1,s_2)\in\mathbb{R}^2$. *Then we have*

$$\begin{aligned} &G_\phi^{A_1,A_2}\{\mathbb{M}_s f\}(\mathbf{w},\mathbf{u})\\ &\quad= e^{\mathbf{i}\omega_1 s_1 d_1} e^{-\mathbf{i}\frac{b_1 d_1 s_1^2}{2}} G_\phi^{A_1,A_2}\{f\}(\mathbf{v},\mathbf{u}) e^{\mathbf{j}\omega_2 s_2 d_2} e^{-\mathbf{j}\frac{b_2 d_2 s_2^2}{2}}, \end{aligned} \tag{3.17}$$

where $\mathbf{v}=(v_1,v_2)\in\mathbb{R}^2$, $v_i=w_i-s_i b_i$, $i=1,2$.

Proof. From Definition 3.1, it follows that

$$\begin{aligned} &G_\phi^{A_1,A_2}\{\mathbb{M}_s f\}(\mathbf{w},\mathbf{u})\\ &\quad=\int_{\mathbb{R}^2}\frac{1}{\sqrt{2\pi b_1}} e^{\mathbf{i}\left(\frac{a_1}{2b_1}x_1^2-\frac{x_1\omega_1}{b_1}+\frac{d_1}{2b_1}\omega_1^2-\frac{\pi}{4}\right)} e^{\mathbf{i}x_1 s_1} f(\mathbf{x})\overline{\phi(\mathbf{x}-\mathbf{u})}\\ &\qquad\times e^{\mathbf{j}x_2 s_2}\frac{1}{\sqrt{2\pi b_2}} e^{\mathbf{j}\left(\frac{a_2}{2b_2}x_2^2-\frac{x_2\omega_2}{b_2}+\frac{d_2}{2b_2}\omega_2^2-\frac{\pi}{4}\right)}\mathrm{d}\mathbf{x}\\ &\quad=\frac{1}{\sqrt{2\pi b_1}}\int_{\mathbb{R}^2} e^{\mathbf{i}\left(\frac{a_1}{2b_1}x_1^2-\frac{x_1(\omega_1-s_1b_1)}{b_1}+\frac{d_1}{2b_1}((\omega_1-s_1b_1)+s_1b_1)^2-\frac{\pi}{4}\right)}\\ &\qquad\times f(\mathbf{x})\overline{\phi(\mathbf{x}-\mathbf{u})}\frac{1}{\sqrt{2\pi b_2}} e^{\mathbf{j}\left(\frac{a_2}{2b_2}x_2^2-\frac{x_2(\omega_2-s_2b_2)}{b_2}+\frac{d_2}{2b_2}((\omega_2-s_2b_2)+s_2b_2)^2-\frac{\pi}{4}\right)}\mathrm{d}\mathbf{x}\\ &\quad= e^{\mathbf{i}\omega_1 s_1 d_1} e^{-\mathbf{i}\frac{b_1 d_1 s_1^2}{2}}\int_{\mathbb{R}^2}\frac{1}{\sqrt{2\pi b_1}} e^{\mathbf{i}\left(\frac{a_1}{2b_1}x_1^2-\frac{x_1(\omega_1-s_1b_1)}{b_1}+\frac{d_1}{2b_1}(\omega_1-s_1b_1)^2-\frac{\pi}{4}\right)} f(\mathbf{x})\\ &\qquad\times\overline{\phi(\mathbf{x}-\mathbf{u})} e^{\mathbf{j}\omega_2 s_2 d_2} e^{-\mathbf{j}\frac{b_2 d_2 s_2^2}{2}}\frac{1}{\sqrt{2\pi b_2}} e^{\mathbf{j}\left(\frac{a_2}{2b_2}x_2^2-\frac{x_2(\omega_2-s_2b_2)}{b_2}+\frac{d_2}{2b_2}(\omega_2-s_2b_2)^2-\frac{\pi}{4}\right)}\mathrm{d}\mathbf{x}. \end{aligned} \tag{3.18}$$

Hence,

$$G_\phi^{A_1,A_2}\{\mathbb{M}_s f\}(\mathbf{w},\mathbf{u}) = e^{\mathbf{i}\omega_1 s_1 d_1} e^{-\mathbf{i}\frac{b_1 d_1 s_1^2}{2}} G_\phi^{A_1,A_2}\{f\}(\mathbf{v},\mathbf{u}) e^{\mathbf{j}\omega_2 s_2 d_2} e^{-\mathbf{j}\frac{b_2 d_2 s_2^2}{2}}, \tag{3.19}$$

which completes the proof. $\square$

Theorem 3.7. (Inversion formula) *Let* $\phi\in L^2(\mathbb{R}^2,\mathbb{H})\backslash\{0\}$ *be a window function,* $0<\|\phi\|^2<\infty$ *and* $f\in L^2\left(\mathbb{R}^2,\mathbb{H}\right)$. *Then we have the inversion formula of the QWLCT,*

$$\begin{aligned} f(\mathbf{x}) = \frac{1}{\|\phi\|^2}\int_{\mathbb{R}^2}\int_{\mathbb{R}^2} K_{A_1}^{-\mathbf{i}(x_1,\omega_1)} G_\phi^{A_1,A_2}\{f\}(\mathbf{w},\mathbf{u}) K_{A_2}^{-\mathbf{j}(x_2,\omega_2)}\phi(\mathbf{x}\\ -\mathbf{u})\mathrm{d}\mathbf{w}\mathrm{d}\mathbf{u}. \end{aligned} \tag{3.20}$$

Proof. Multiplying both sides of (3.3) from the right by $\phi(\mathbf{x}-\mathbf{u})$ and integrating with respect to $\mathrm{d}\mathbf{u}$ we get

$$\begin{aligned} \int_{\mathbb{R}^2} f(\mathbf{x})\overline{\phi(\mathbf{x}-\mathbf{u})}\phi(\mathbf{x}-\mathbf{u})\mathrm{d}\mathbf{u} = \int_{\mathbb{R}^2}\int_{\mathbb{R}^2} K_{A_1}^{-\mathbf{i}(x_1,\omega_1)} G_\phi^{A_1,A_2}\{f\}(\mathbf{w},\mathbf{u})\\ \times K_{A_2}^{-\mathbf{j}(x_2,\omega_2)}\phi(\mathbf{x}-\mathbf{u})\mathrm{d}\mathbf{w}\mathrm{d}\mathbf{u}. \end{aligned} \tag{3.21}$$

Using (2.9), we have

$$f(\mathbf{x}) = \frac{1}{\|\phi\|^2}\int_{\mathbb{R}^2}\int_{\mathbb{R}^2} K_{A_1}^{-\mathbf{i}(x_1,\omega_1)} G_\phi^{A_1,A_2}\{f\}(\mathbf{w},\mathbf{u}) K_{A_2}^{-\mathbf{j}(x_2,\omega_2)}\phi(\mathbf{x} -\mathbf{u})\mathrm{d}\mathbf{w}\mathrm{d}\mathbf{u}, \tag{3.22}$$

which completes the proof. □

Theorem 3.8. (Orthogonality relation) *Let $\phi,\psi \in L^2(\mathbb{R}^2,\mathbb{H})\backslash\{0\}$ be window functions and $f,g \in L^2\left(\mathbb{R}^2,\mathbb{H}\right)$. Then*

$$\langle G_\phi^{A_1,A_2}\{f\}(\mathbf{w},\mathbf{u}), G_\psi^{A_1,A_2}\{g\}(\mathbf{w},\mathbf{u})\rangle = [(f,g)(\phi,\psi)]_0. \tag{3.23}$$

Proof. Using (2.3), (2.8), we get the output as follows:

$$\begin{aligned}
&\langle G_\phi^{A_1,A_2}\{f\}(\mathbf{w},\mathbf{u}), G_\psi^{A_1,A_2}\{g\}(\mathbf{w},\mathbf{u})\rangle \\
&= \int_{\mathbb{R}^4}[G_\phi^{A_1,A_2}\{f\}(\mathbf{w},\mathbf{u})\overline{G_\psi^{A_1,A_2}\{g\}(\mathbf{w},\mathbf{u})}]_0\mathrm{d}\mathbf{w}\mathrm{d}\mathbf{u} \\
&= \int_{\mathbb{R}^4}[G_\phi^{A_1,A_2}\{f\}(\mathbf{w},\mathbf{u}) \\
&\quad\times\overline{\int_{\mathbb{R}^2} K_{A_1}^{\mathbf{i}}(x_1,\omega_1)g(\mathbf{x})\overline{\psi(\mathbf{x}-\mathbf{u})}K_{A_2}^{\mathbf{j}}(x_2,\omega_2)\mathrm{d}\mathbf{x}}]_0\mathrm{d}\mathbf{w}\mathrm{d}\mathbf{u} \\
&= \int_{\mathbb{R}^6}[G_\phi^{A_1,A_2}\{f\}(\mathbf{w},\mathbf{u})K_{A_2}^{-\mathbf{j}}(x_2,\omega_2)\psi(\mathbf{x}-\mathbf{u})\overline{g(\mathbf{x})}K_{A_1}^{-\mathbf{i}}(x_1,\omega_1)]_0\mathrm{d}\mathbf{w}\mathrm{d}\mathbf{u}\mathrm{d}\mathbf{x} \\
&= \int_{\mathbb{R}^6}[K_{A_1}^{-\mathbf{i}}(x_1,\omega_1)G_\phi^{A_1,A_2}\{f\}(\mathbf{w},\mathbf{u})K_{A_2}^{-\mathbf{j}}(x_2,\omega_2)\psi(\mathbf{x}-\mathbf{u})\overline{g(\mathbf{x})}]_0\mathrm{d}\mathbf{w}\mathrm{d}\mathbf{u}\mathrm{d}\mathbf{x} \\
&= \int_{\mathbb{R}^4}\left[\int_{\mathbb{R}^2} K_{A_1}^{-\mathbf{i}}(x_1,\omega_1)G_\phi^{A_1,A_2}\{f\}(\mathbf{w},\mathbf{u})K_{A_2}^{-\mathbf{j}}(x_2,\omega_2)\mathrm{d}\mathbf{w}\psi(\mathbf{x}-\mathbf{u})\overline{g(\mathbf{x})}\right]_0\mathrm{d}\mathbf{u}\mathrm{d}\mathbf{x}.
\end{aligned} \tag{3.24}$$

Because

$$\overline{K_{A_1}^{\mathbf{i}}(x_1,\omega_1)} = K_{A_1}^{-\mathbf{i}}(x_1,\omega_1) = K_{A_1^{-1}}^{\mathbf{i}}(\omega_1,x_1), \tag{3.25}$$

$$\overline{K_{A_2}^{\mathbf{j}}(x_2,\omega_2)} = K_{A_2}^{-\mathbf{j}}(x_2,\omega_2) = K_{A_2^{-1}}^{\mathbf{j}}(\omega_2,x_2). \tag{3.26}$$

Using (3.3), we have

$$\begin{aligned}
&\langle G_\phi^{A_1,A_2}\{f\}(\mathbf{w},\mathbf{u}), G_\psi^{A_1,A_2}\{g\}(\mathbf{w},\mathbf{u})\rangle \\
&\quad= \int_{\mathbb{R}^4}\left[\int_{\mathbb{R}^2} K_{A_1^{-1}}^{\mathbf{i}}(\omega_1,x_1)G_\phi^{A_1,A_2}\{f\}(\mathbf{w},\mathbf{u})K_{A_2^{-1}}^{\mathbf{j}}(\omega_2,x_2)\mathrm{d}\mathbf{w}\right. \\
&\quad\quad\left.\times\psi(\mathbf{x}-\mathbf{u})\overline{g(\mathbf{x})}\right]_0\mathrm{d}\mathbf{u}\mathrm{d}\mathbf{x} \\
&\quad= \int_{\mathbb{R}^4}\left[f(\mathbf{x})\overline{\phi(\mathbf{x}-\mathbf{u})}\psi(\mathbf{x}-\mathbf{u})\overline{g(\mathbf{x})}\right]_0\mathrm{d}\mathbf{u}\mathrm{d}\mathbf{x}
\end{aligned} \tag{3.27}$$

Using the change of variables $\mathbf{x} - \mathbf{u} = \mathbf{y}$, the equation becomes

$$\begin{aligned} \langle G_{\phi}^{A_1,A_2}\{f\}(\mathbf{w},\mathbf{u}), G_{\psi}^{A_1,A_2}\{g\}(\mathbf{w},\mathbf{u})\rangle &= \int_{\mathbb{R}^4} \left[f(\mathbf{x})\overline{\phi(\mathbf{y})}\psi(\mathbf{y})\overline{g(\mathbf{x})}\right]_0 \mathrm{d}\mathbf{y}\mathrm{d}\mathbf{x} \\ &= \left[\int_{\mathbb{R}^2} f(\mathbf{x})\overline{g(\mathbf{x})}\mathrm{d}\mathbf{x} \int_{\mathbb{R}^2} \psi(\mathbf{y})\overline{\phi(\mathbf{y})}\mathrm{d}\mathbf{y}\right]_0 \\ &= [(f,g)(\phi,\psi)]_0, \end{aligned} \tag{3.28}$$

which completes the proof. $\square$

Based on the above theorem, we may conclude the following important consequences (Table 1).

(i) If $\phi = \psi$, then

$$\langle G_{\phi}^{A_1,A_2}\{f\}(\mathbf{w},\mathbf{u}), G_{\phi}^{A_1,A_2}\{g\}(\mathbf{w},\mathbf{u})\rangle = \|\phi\|^2_{L^2(\mathbb{R}^2)}\langle f, g\rangle \tag{3.29}$$

(ii) If $f = g$, then

$$\langle G_{\phi}^{A_1,A_2}\{f\}(\mathbf{w},\mathbf{u}), G_{\psi}^{A_1,A_2}\{f\}(\mathbf{w},\mathbf{u})\rangle = \|f\|^2_{L^2(\mathbb{R}^2)}\langle \phi, \psi\rangle \tag{3.30}$$

(iii) If $f = g$ and $\phi = \psi$, then

$$\begin{aligned} \langle G_{\phi}^{A_1,A_2}\{f\}(\mathbf{w},\mathbf{u}), G_{\phi}^{A_1,A_2}\{f\}(\mathbf{w},\mathbf{u})\rangle &= \int_{\mathbb{R}^2}\int_{\mathbb{R}^2} |G_{\phi}^{A_1,A_2}\{f\}(\mathbf{w},\mathbf{u})|^2 \mathrm{d}\mathbf{w}\mathrm{d}\mathbf{u} \\ &= \|f\|^2_{L^2(\mathbb{R}^2)}\|\phi\|^2_{L^2(\mathbb{R}^2)}. \end{aligned} \tag{3.31}$$

4. Heisenberg's Uncertainty Principle for the QWLCT

The Heisenberg uncertainty principle and quaternions are both basic for quantum mechanics. In quantum mechanics an uncertainty principle asserts that one cannot make certain of the position and velocity of an electron (or any particle) at the same time. That is, increasing the knowledge of the position decreases the knowledge of the velocity or momentum of an electron. In signal processing an uncertainty principle states that the product of the variances of the signal in the time and frequency domains has a lower bound. The uncertainty principles of the Fourier transform and the quaternion Fourier transform had been studied in Refs. [3, 29]. Recently, the authors established the uncertainty principles associated with the LCT in Refs. [9, 10, 20, 25, 27, 31]. The uncertainty principles for the WLCT had been discussed in Refs. [17]. Theirs uncertainties were generalizations of Lieb's uncertainty principles in the WLCT domains. Recently, Heisenberg's uncertainty relations were extended to the quaternion linear canonical transform [14, 23]. This uncertainty principle prescribes a lower bound on the product of the effective widths of quaternion-valued signals in the spatial and frequency domains. Let us give a short and simple proof of the Heisenberg's Uncertainty Principle of the QWLCT.

Lemma 4.1. (QLCT uncertainty principle) *[14] Let $f \in L^2\left(\mathbb{R}^2, \mathbb{H}\right)$ and*

TABLE 1. Properties of the QWLCT of $\phi \in L^2(\mathbb{R}^2, \mathbb{H})\backslash\{0\}$ and $f, g \in L^2\left(\mathbb{R}^2, \mathbb{H}\right)$, where $\lambda, \mu \in \mathbb{H}$ are arbitrary constants

Property	Function	QWLCT
Boundedness	$\lvert G_{\phi}^{A_1,A_2} f(\mathbf{w}, \mathbf{u})\rvert \leq$	$\frac{1}{2\pi\sqrt{\lvert b_1 b_2\rvert}} \lVert f\rVert_{L^2(\mathbb{R}^2)} \lVert\phi\rVert_{L^2(\mathbb{R}^2)}$
Linearity	$\lambda f + \mu g$	$\lambda G_{\phi}^{A_1,A_2} f(\mathbf{w}, \mathbf{u}) + \mu G_{\phi}^{A_1,A_2} g(\mathbf{w}, \mathbf{u})$
Parity	$G_{P\phi}^{A_1,A_2}\{Pf\}(\mathbf{w}, \mathbf{u})$	$G_{\phi}^{A_1,A_2} f(-\mathbf{w}, -\mathbf{u})$
Shift	$f(\mathbf{x} - \mathbf{r})$	$e^{\mathbf{i} r_1 \omega_1 c_1} e^{-\mathbf{i}\frac{a_1 r_1^2}{2} c_1} G_{\phi}^{A_1,A_2}\{f\}(\mathbf{m}, \mathbf{n})$ $\times e^{\mathbf{j} r_2 \omega_2 c_2} e^{-\mathbf{j}\frac{a_2 r_2^2}{2} c_2}$
Modulation	$e^{\mathbf{i} x_1 s_1} f(\mathbf{x}) e^{\mathbf{j} x_2 s_2}$	$e^{\mathbf{i}\omega_1 s_1 d_1} e^{-\mathbf{i}\frac{b_1 d_1 s_1^2}{2}} G_{\phi}^{A_1,A_2}\{f\}(\mathbf{v}, \mathbf{u})$ $\times e^{\mathbf{j}\omega_2 s_2 d_2} e^{-\mathbf{j}\frac{b_2 d_2 s_2^2}{2}}$
Formula		
Inversion	$f(\mathbf{x}) = \frac{1}{\lVert\phi\rVert^2} \int_{\mathbb{R}^2}\int_{\mathbb{R}^2} \overline{K_{A_1}^{\mathbf{i}}(x_1, \omega_1)} G_{\phi}^{A_1,A_2}\{f\}(\mathbf{w}, \mathbf{u})$ $\times \overline{K_{A_2}^{\mathbf{j}}(x_2, \omega_2)} \phi(\mathbf{x} - \mathbf{u}) \mathrm{d}\mathbf{w}\mathrm{d}\mathbf{u}$	
Orthogonality	$\langle G_{\phi}^{A_1,A_2}\{f\}(\mathbf{w}, \mathbf{u}), G_{\psi}^{A_1,A_2}\{g\}(\mathbf{w}, \mathbf{u})\rangle = [\langle f, g\rangle\langle\phi, \psi\rangle]_0$ $\langle G_{\phi}^{A_1,A_2}\{f\}(\mathbf{w}, \mathbf{u}), G_{\phi}^{A_1,A_2}\{g\}(\mathbf{w}, \mathbf{u})\rangle = \lVert\phi\rVert_{L^2(\mathbb{R}^2)}^2 \langle f, g\rangle$ $\langle G_{\phi}^{A_1,A_2}\{f\}(\mathbf{w}, \mathbf{u}), G_{\psi}^{A_1,A_2}\{f\}(\mathbf{w}, \mathbf{u})\rangle = \lVert f\rVert_{L^2(\mathbb{R}^2)}^2 \langle\phi, \psi\rangle$ $\langle G_{\phi}^{A_1,A_2}\{f\}(\mathbf{w}, \mathbf{u}), G_{\phi}^{A_1,A_2}\{f\}(\mathbf{w}, \mathbf{u})\rangle$ $= \int_{\mathbb{R}^2}\int_{\mathbb{R}^2} \lvert G_{\phi}^{A_1,A_2}\{f\}(\mathbf{w}, \mathbf{u})\rvert^2 \mathrm{d}\mathbf{w}\mathrm{d}\mathbf{u} = \lVert f\rVert_{L^2(\mathbb{R}^2)}^2 \lVert\phi\rVert_{L^2(\mathbb{R}^2)}^2$	

$\mathcal{L}_{A_1,A_2}^{\mathbb{H}}\{f\}(\mathbf{w}) \in L^2\left(\mathbb{R}^2, \mathbb{H}\right)$ *then we have*

$$\int_{\mathbb{R}^2} x_k^2 |f(\mathbf{x})|^2 \mathrm{d}\mathbf{x} \int_{\mathbb{R}^2} \omega_k^2 |\mathcal{L}_{A_1,A_2}^{\mathbb{H}}\{f\}(\mathbf{w})|^2 \mathrm{d}\mathbf{w} \geq \frac{b_k^2}{4}\left(\int_{\mathbb{R}^2} |f(\mathbf{x})|^2 \mathrm{d}\mathbf{x}\right)^2, \tag{4.1}$$

where $k = 1, 2$.

Substituting the inverse transform for the QLCT (2.14) into the left-hand side of (4.1), we obtain

$$\begin{aligned} &\int_{\mathbb{R}^2} x_k^2 |\mathcal{L}_{A_1,A_2}^{-1}[\mathcal{L}_{A_1,A_2}^{\mathbb{H}}\{f\}]|^2 \mathrm{d}\mathbf{x} \int_{\mathbb{R}^2} \omega_k^2 |\mathcal{L}_{A_1,A_2}^{\mathbb{H}}\{f\}(\mathbf{w})|^2 \mathrm{d}\mathbf{w} \\ &\quad \geq \frac{b_k^2}{4}\left(\int_{\mathbb{R}^2} |f(\mathbf{x})|^2 \mathrm{d}\mathbf{x}\right)^2, \end{aligned} \tag{4.2}$$

where $k=1,2$. Further, applying Plancherel's theorem for the QLCT (2.16) to the right-hand side of (4.2), we have

$$\int_{\mathbb{R}^2} x_k^2 |\mathcal{L}^{-1}_{A_1,A_2}[\mathcal{L}^{\mathbb{H}}_{A_1,A_2}\{f\}]|^2 \mathrm{d}\mathbf{x} \int_{\mathbb{R}^2} \omega_k^2 |\mathcal{L}^{\mathbb{H}}_{A_1,A_2}\{f\}(\mathbf{w})|^2 \mathrm{d}\mathbf{w} \\ \geq \frac{b_k^2}{4}\left(\int_{\mathbb{R}^2} |\mathcal{L}^{\mathbb{H}}_{A_1,A_2}\{f\}(\mathbf{w})|^2 \mathrm{d}\mathbf{w}\right)^2, k=1,2. \tag{4.3}$$

Lemma 4.2. *Let $\phi \in L^2(\mathbb{R}^2,\mathbb{H})\backslash\{0\}$ be a window function and $f \in L^2\left(\mathbb{R}^2,\mathbb{H}\right)$. Then*

$$\|\phi\|^2_{L^2(\mathbb{R}^2)} \int_{\mathbb{R}^2} x_k^2 |f(\mathbf{x})|^2 \mathrm{d}\mathbf{x} \\ = \int_{\mathbb{R}^2}\int_{\mathbb{R}^2} x_k^2 |\mathcal{L}^{-1}_{A_1,A_2}\{G_\phi^{A_1,A_2} f(\mathbf{w},\mathbf{u})\}(\mathbf{x})|^2 \mathrm{d}\mathbf{x}\mathrm{d}\mathbf{u}. \tag{4.4}$$

Proof. Based on the result of Lemma 3.4 from [5], we get

$$\begin{aligned} \|\phi\|^2_{L^2(\mathbb{R}^2)} \int_{\mathbb{R}^2} x_k^2 |f(\mathbf{x})|^2 \mathrm{d}\mathbf{x} &= \int_{\mathbb{R}^2} x_k^2 |f(\mathbf{x})|^2 \mathrm{d}\mathbf{x} \int_{\mathbb{R}^2} |\phi(\mathbf{x}-\mathbf{u})|^2 \mathrm{d}\mathbf{u} \\ &= \int_{\mathbb{R}^2}\int_{\mathbb{R}^2} x_k^2 |f(\mathbf{x})|^2 |\phi(\mathbf{x}-\mathbf{u})|^2 \mathrm{d}\mathbf{x}\mathrm{d}\mathbf{u} \\ &= \int_{\mathbb{R}^2}\int_{\mathbb{R}^2} x_k^2 |f(\mathbf{x})\overline{\phi(\mathbf{x}-\mathbf{u})}|^2 \mathrm{d}\mathbf{x}\mathrm{d}\mathbf{u} \\ &= \int_{\mathbb{R}^2}\int_{\mathbb{R}^2} x_k^2 |\mathcal{L}^{-1}_{A_1,A_2}\{G_\phi^{A_1,A_2} f(\mathbf{w},\mathbf{u})\}(\mathbf{x})|^2 \mathrm{d}\mathbf{x}\mathrm{d}\mathbf{u} \end{aligned} \tag{4.5}$$

□

Now we arrive at the following important result.

Theorem 4.3. (QWLCT uncertainty principle) *Let $\phi \in L^2(\mathbb{R}^2,\mathbb{H})\backslash\{0\}$ be a window function and $G_\phi^{A_1,A_2}\{f\} \in L^2\left(\mathbb{R}^2,\mathbb{H}\right)$ be the QWLCT of f. Then for every $f \in L^2\left(\mathbb{R}^2,\mathbb{H}\right)$ we have the following inequality*

$$\left(\int_{\mathbb{R}^2}\int_{\mathbb{R}^2} \omega_k^2 |G_\phi^{A_1,A_2} f(\mathbf{w},\mathbf{u})|^2 \mathrm{d}\mathbf{w}\mathrm{d}\mathbf{u}\right)^{\frac{1}{2}} \left(\int_{\mathbb{R}^2} x_k^2 |f(\mathbf{x})|^2 \mathrm{d}\mathbf{x}\right)^{\frac{1}{2}} \\ \geq \frac{b_k}{2}\|f\|^2_{L^2(\mathbb{R}^2)}\|\phi\|_{L^2(\mathbb{R}^2)}. \tag{4.6}$$

Proof. Assume that $\mathcal{L}^{\mathbb{H}}_{A_1,A_2}\{f\} \in L^2\left(\mathbb{R}^2,\mathbb{H}\right)$. Since $G_\phi^{A_1,A_2} f \in L^2\left(\mathbb{R}^2,\mathbb{H}\right)$ we can replace the QLCT of f by the QWLCT of f on the both sides of (4.3). We have

$$\int_{\mathbb{R}^2} x_k^2 |\mathcal{L}^{-1}_{A_1,A_2}[G_\phi^{A_1,A_2} f(\mathbf{w},\mathbf{u})]|^2 \mathrm{d}\mathbf{x} \int_{\mathbb{R}^2} \omega_k^2 |G_\phi^{A_1,A_2} f(\mathbf{w},\mathbf{u})|^2 \mathrm{d}\mathbf{w} \\ \geq \frac{b_k^2}{4}\left(\int_{\mathbb{R}^2} |G_\phi^{A_1,A_2} f(\mathbf{w},\mathbf{u})|^2 \mathrm{d}\mathbf{w}\right)^2. \tag{4.7}$$

Taking the square root on both sides of (4.7) and integrating both sides with respect to d$\mathbf{u}$, we have

$$\begin{aligned}\int_{\mathbb{R}^2}\left(\int_{\mathbb{R}^2} x_k^2|\mathcal{L}^{-1}_{A_1,A_2}[G_\phi^{A_1,A_2}f(\mathbf{w},\mathbf{u})]|^2\mathrm{d}\mathbf{x}\right)^{\frac{1}{2}}\\ \times\left(\int_{\mathbb{R}^2}\omega_k^2|G_\phi^{A_1,A_2}f(\mathbf{w},\mathbf{u})|^2\mathrm{d}\mathbf{w}\right)^{\frac{1}{2}}\mathrm{d}\mathbf{u}\\ \geq \frac{b_k}{2}\int_{\mathbb{R}^2}\int_{\mathbb{R}^2}|G_\phi^{A_1,A_2}f(\mathbf{w},\mathbf{u})|^2\mathrm{d}\mathbf{w}\mathrm{d}\mathbf{u}\end{aligned} \tag{4.8}$$

Furthermore, applying the Cauchy–Schwarz inequality to the left-hand side of (4.8), we have

$$\begin{aligned}\left(\int_{\mathbb{R}^2}\int_{\mathbb{R}^2} x_k^2|\mathcal{L}^{-1}_{A_1,A_2}[G_\phi^{A_1,A_2}f(\mathbf{w},\mathbf{u})]|^2\mathrm{d}\mathbf{x}\mathrm{d}\mathbf{u}\right)^{\frac{1}{2}}\\ \times\left(\int_{\mathbb{R}^2}\int_{\mathbb{R}^2}\omega_k^2|G_\phi^{A_1,A_2}f(\mathbf{w},\mathbf{u})|^2\mathrm{d}\mathbf{w}\mathrm{d}\mathbf{u}\right)^{\frac{1}{2}}\\ \geq \frac{b_k}{2}\int_{\mathbb{R}^2}\int_{\mathbb{R}^2}|G_\phi^{A_1,A_2}f(\mathbf{w},\mathbf{u})|^2\mathrm{d}\mathbf{w}\mathrm{d}\mathbf{u}\end{aligned} \tag{4.9}$$

Inserting (4.4) into the second term on the left-hand side of (4.9) and substituting (3.31) into the right-hand side of this inequality. We have

$$\begin{aligned}\left(\|\phi\|^2_{L^2(\mathbb{R}^2)}\int_{\mathbb{R}^2}x_k^2|f(\mathbf{x})|^2\mathrm{d}\mathbf{x}\right)^{\frac{1}{2}}\left(\int_{\mathbb{R}^2}\int_{\mathbb{R}^2}\omega_k^2|G_\phi^{A_1,A_2}f(\mathbf{w},\mathbf{u})|^2\mathrm{d}\mathbf{w}\mathrm{d}\mathbf{u}\right)^{\frac{1}{2}}\\ \geq \frac{b_k}{2}\|f\|^2_{L^2(\mathbb{R}^2)}\|\phi\|^2_{L^2(\mathbb{R}^2)}\end{aligned} \tag{4.10}$$

Dividing both sides of (4.10) by $\|\phi\|_{L^2(\mathbb{R}^2)}$, we obtain the desired result. □

5. Example of the QWLCT

Given the window function of the two-dimensional Haar function defined by

$$\phi(\mathbf{x}) = \begin{cases} 1, & 0 \leq x_1 < \frac{1}{2}, 0 \leq x_2 < \frac{1}{2} \\ -1, & \frac{1}{2} \leq x_1 < 1, \frac{1}{2} \leq x_2 < 1 \\ 0, & otherwise. \end{cases} \tag{5.1}$$

Consider a 2D Gaussian quaternionic function of the form $f(x_1, x_2) = e^{-(\alpha_1 x_1^2 + \alpha_2 x_2^2)}$, for $\alpha_1, \alpha_2 \in \mathbb{R}$ are positive real constants.

Then the QWLCT of f is given by

$$\begin{aligned}
G_{\phi}^{A_1,A_2} f(\mathbf{w},\mathbf{u}) &= \int_{\mathbb{R}^2} \frac{1}{\sqrt{2\pi b_1}} e^{\mathbf{i}\left(\frac{a_1}{2b_1}x_1^2 - \frac{x_1 u_1}{b_1} + \frac{d_1}{2b_1}u_1^2 - \frac{\pi}{4}\right)} f(\mathbf{x})\overline{\phi(\mathbf{x}-\mathbf{u})} \\
&\quad \times \frac{1}{\sqrt{2\pi b_2}} e^{\mathbf{j}\left(\frac{a_2}{2b_2}x_2^2 - \frac{x_2 u_2}{b_2} + \frac{d_2}{2b_2}u_2^2 - \frac{\pi}{4}\right)} \mathrm{d}\mathbf{x} \\
&= \int_{u_1}^{\frac{1}{2}+u_1} \frac{1}{\sqrt{2\pi b_1}} e^{\mathbf{i}\left(\frac{a_1}{2b_1}x_1^2 - \frac{x_1 u_1}{b_1} + \frac{d_1}{2b_1}u_1^2 - \frac{\pi}{4}\right)} e^{-\alpha_1 x_1^2} \mathrm{d}x_1 \\
&\quad \times \int_{u_2}^{\frac{1}{2}+u_2} \frac{1}{\sqrt{2\pi b_2}} e^{\mathbf{j}\left(\frac{a_2}{2b_2}x_2^2 - \frac{x_2 u_2}{b_2} + \frac{d_2}{2b_2}u_2^2 - \frac{\pi}{4}\right)} e^{-\alpha_2 x_2^2} \mathrm{d}x_2 \\
&\quad - \int_{\frac{1}{2}+u_1}^{1+u_1} \frac{1}{\sqrt{2\pi b_1}} e^{\mathbf{i}\left(\frac{a_1}{2b_1}x_1^2 - \frac{x_1 u_1}{b_1} + \frac{d_1}{2b_1}u_1^2 - \frac{\pi}{4}\right)} e^{-\alpha_1 x_1^2} \mathrm{d}x_1 \\
&\quad \times \int_{\frac{1}{2}+u_2}^{1+u_2} \frac{1}{\sqrt{2\pi b_2}} e^{\mathbf{j}\left(\frac{a_2}{2b_2}x_2^2 - \frac{x_2 u_2}{b_2} + \frac{d_2}{2b_2}u_2^2 - \frac{\pi}{4}\right)} e^{-\alpha_2 x_2^2} \mathrm{d}x_2.
\end{aligned} \tag{5.2}$$

By completing squares, we have

$$\begin{aligned}
&G_{\phi}^{A_1,A_2} f(\mathbf{w},\mathbf{u}) \\
&= \int_{u_1}^{\frac{1}{2}+u_1} \frac{1}{\sqrt{2\pi b_1}} e^{-\left(\sqrt{\frac{2b_1\alpha_1 - \mathbf{i}a_1}{2b_1}}x_1 + \frac{\mathbf{i}\omega_1}{\sqrt{2b_1(2b_1\alpha_1-\mathbf{i}a_1)}}\right)^2} e^{-\frac{\omega_1^2}{2b_1\alpha_1-\mathbf{i}a_1} + \frac{\mathbf{i}d_1}{2b_1}\omega_1^2 - \frac{\pi}{4}} \mathrm{d}x_1 \\
&\quad \times \int_{u_2}^{\frac{1}{2}+u_2} \frac{1}{\sqrt{2\pi b_2}} e^{-\left(\sqrt{\frac{2b_2\alpha_2 - \mathbf{j}a_2}{2b_2}}x_2 + \frac{\mathbf{j}\omega_2}{\sqrt{2b_2(2b_2\alpha_2-\mathbf{j}a_2)}}\right)^2} e^{-\frac{\omega_2^2}{2b_2\alpha_2-\mathbf{j}a_2} + \frac{\mathbf{j}d_2}{2b_2}\omega_2^2 - \frac{\pi}{4}} \mathrm{d}x_2 \\
&\quad - \int_{\frac{1}{2}+u_1}^{1+u_1} \frac{1}{\sqrt{2\pi b_1}} e^{-\left(\sqrt{\frac{2b_1\alpha_1 - \mathbf{i}a_1}{2b_1}}x_1 + \frac{\mathbf{i}\omega_1}{\sqrt{2b_1(2b_1\alpha_1-\mathbf{i}a_1)}}\right)^2} e^{-\frac{\omega_1^2}{2b_1\alpha_1-\mathbf{i}a_1} + \frac{\mathbf{i}d_1}{2b_1}\omega_1^2 - \frac{\pi}{4}} \mathrm{d}x_1 \\
&\quad \times \int_{\frac{1}{2}+u_2}^{1+u_2} \frac{1}{\sqrt{2\pi b_2}} e^{-\left(\sqrt{\frac{2b_2\alpha_2 - \mathbf{j}a_2}{2b_2}}x_2 + \frac{\mathbf{j}\omega_2}{\sqrt{2b_2(2b_2\alpha_2-\mathbf{j}a_2)}}\right)^2} e^{-\frac{\omega_2^2}{2b_2\alpha_2-\mathbf{j}a_2} + \frac{\mathbf{j}d_2}{2b_2}\omega_2^2 - \frac{\pi}{4}} \mathrm{d}x_2.
\end{aligned} \tag{5.3}$$

Making the substitutions $A_1 = \sqrt{\frac{2b_1\alpha_1 - \mathbf{i}a_1}{2b_1}}$, $A_2 = \sqrt{\frac{2b_2\alpha_2 - \mathbf{j}a_2}{2b_2}}$, $B_1 = \frac{\mathbf{i}\omega_1}{\sqrt{2b_1(2b_1\alpha_1-\mathbf{i}a_1)}}$, $B_2 = \frac{\mathbf{j}\omega_2}{\sqrt{2b_2(2b_2\alpha_2-\mathbf{j}a_2)}}$, $J_1 = e^{-\frac{\omega_1^2}{2b_1\alpha_1-\mathbf{i}a_1} + \frac{\mathbf{i}d_1}{2b_1}\omega_1^2 - \frac{\pi}{4}}$ and $J_2 = e^{-\frac{\omega_2^2}{2b_2\alpha_2-\mathbf{j}a_2} + \frac{\mathbf{j}d_2}{2b_2}\omega_2^2 - \frac{\pi}{4}}$ in the above expression we immediately obtain

$$\begin{aligned}
&G_{\phi}^{A_1,A_2} f(\mathbf{w},\mathbf{u}) \\
&= \int_{u_1}^{\frac{1}{2}+u_1} \frac{1}{\sqrt{2\pi b_1}} e^{-(A_1x_1+B_1)^2} J_1 \mathrm{d}x_1 \int_{u_2}^{\frac{1}{2}+u_2} \frac{1}{\sqrt{2\pi b_2}} e^{-(A_2x_2+B_2)^2} J_2 \mathrm{d}x_2 \\
&\quad - \int_{\frac{1}{2}+u_1}^{1+u_1} \frac{1}{\sqrt{2\pi b_1}} e^{-(A_1x_1+B_1)^2} J_1 \mathrm{d}x_1 \int_{\frac{1}{2}+u_2}^{1+u_2} \frac{1}{\sqrt{2\pi b_2}} e^{-(A_2x_2+B_2)^2} J_2 \mathrm{d}x_2.
\end{aligned} \tag{5.4}$$

Substituting $y_1 = A_1x_1+B_1$ and $y_2 = A_2x_2+B_2$ in the above equation, we have

$$\begin{aligned}
&G_\phi^{A_1,A_2} f(\mathbf{w},\mathbf{u})\\
&= \frac{J_1}{A_1\sqrt{2\pi b_1}} \int_{A_1u_1+B_1}^{A_1(\frac{1}{2}+u_1)+B_1} e^{-y_1^2}\mathrm{d}y_1 \times \frac{J_2}{A_2\sqrt{2\pi b_2}} \int_{A_2u_2+B_2}^{A_2(\frac{1}{2}+u_2)+B_2} e^{-y_2^2}\mathrm{d}y_2\\
&\quad -\frac{J_1}{A_1\sqrt{2\pi b_1}} \int_{A_1(\frac{1}{2}+u_1)+B_1}^{A_1(1+u_1)+B_1} e^{-y_1^2}\mathrm{d}x_1 \times \frac{J_2}{A_2\sqrt{2\pi b_2}} \int_{A_2(\frac{1}{2}+u_2)+B_2}^{A_2(1+u_2)+B_2} e^{-y_2^2}\mathrm{d}x_2\\
&= \frac{J_1}{A_1\sqrt{2\pi b_1}} \left(\int_0^{A_1u_1+B_1} (-e^{-y_1^2})\mathrm{d}y_1 + \int_0^{A_1(\frac{1}{2}+u_1)+B_1} e^{-y_1^2}\mathrm{d}y_1\right)\\
&\quad \times\frac{J_2}{A_2\sqrt{2\pi b_2}} \left(\int_0^{A_2u_2+B_2} (-e^{-y_2^2})\mathrm{d}y_2 + \int_0^{A_2(\frac{1}{2}+u_2)+B_2} e^{-y_2^2}\mathrm{d}y_2\right)\\
&\quad -\frac{J_1}{A_1\sqrt{2\pi b_1}} \left(\int_0^{A_1(\frac{1}{2}+u_1)+B_1} (-e^{-y_1^2})\mathrm{d}y_1 + \int_0^{A_1(1+u_1)+B_1} e^{-y_1^2}\mathrm{d}y_1\right)\\
&\quad \times\frac{J_2}{A_2\sqrt{2\pi b_2}} \times \left(\int_0^{A_2(\frac{1}{2}+u_2)+B_2} (-e^{-y_2^2})\mathrm{d}y_2 + \int_0^{A_2(1+u_2)+B_2} e^{-y_2^2}\mathrm{d}y_2\right).
\end{aligned} \tag{5.5}$$

Equation (5.5) can be written in the form

$$\begin{aligned}
&G_\phi^{A_1,A_2} f(\mathbf{w},\mathbf{u})\\
&= \frac{J_1}{2\sqrt{2b_1\alpha_1-\mathbf{i}\alpha_1}} \left(-erf(A_1u_1+B_1) + erf(A_1(\frac{1}{2}+u_1)+B_1)\right)\\
&\quad \times\frac{J_2}{2\sqrt{2b_2\alpha_2-\mathbf{j}\alpha_2}} \left(-erf(A_2u_2+B_2) + erf(A_2(\frac{1}{2}+u_2)+B_2)\right)\\
&\quad -\frac{J_1}{2\sqrt{2b_1\alpha_1-\mathbf{i}\alpha_1}} \left(-erf(A_1(\frac{1}{2}+u_1)+B_1) + erf(A_1(1+u_1)+B_1)\right)\\
&\quad \times\frac{J_2}{2\sqrt{2b_2\alpha_2-\mathbf{j}\alpha_2}} \left(-erf(A_2(\frac{1}{2}+u_2)+B_2) + erf(A_2(1+u_2)+B_2)\right),
\end{aligned} \tag{5.6}$$

where $erf(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}\mathrm{d}t$.

The WLCT plays a vital role in the analysis of signals [16,32]. The windowed linear canonical kernel provides a unique representation for the signals so that the WLCT is highly effective. It is natural to think whether the QWLCT can also be applied to such problems. For illustrative purposes, we give the example above.

6. Conclusions

Due to the non-commutativity of quaternion multiplication, there are three different types of the QLCT: the left-sided QLCT, the right-sided QLCT, and the two-sided QLCT. Using the basic concepts of quaternion algebra and the two-sided QLCT we introduced the the two-sided QWLCT. Important properties of the QWLCT such as boundedness, linearity, parity, shift, modulation, inversion formula and orthogonality relation were derived. Based on the QWLCT properties and the uncertainty principle for the QLCT, a new uncertainty principle for the QWLCT have been established. The results in this paper are new in the literature. Further investigations on this topic are now under investigation such as the left-sided QWLCT, the right-sided QWLCT and the QWLCT directional uncertainty principle. They will be reported in a forthcoming paper.

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